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Self-Reported Precipitating and Priming Factors for Somnambulism in Adult Sleepwalkers

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ABSTRACT

Several studies have explored how factors that deepen sleep (e.g., sleep deprivation) and factors that fragment sleep (e.g., environmental stimuli) facilitate the occurrence of somnambulist episodes experienced by sleepwalkers in the sleep laboratory. Little is known, however, about the broader range of variables that sleepwalkers perceive as contributing to their episodes in their home environment. We examined self-reported precipitating and priming factors for sleepwalking in a large cohort of 188 adults diagnosed with primary somnambulism using a comprehensive questionnaire. The most frequently endorsed precipitating or priming factors for sleepwalking episodes were psychological stress (95%), bad dreams and nightmares (78%), and sleep deprivation (60%). Factors such as irregular sleep schedules, intense movies, and new sleep environments were also relatively common. In contrast, substance use (e.g., alcohol, caffeine) and physical ailments (e.g., fever, pain) were cited less frequently. We found few significant differences in reported factors based on biological sex, age of onset, or family history, although women were more likely to associate noisy environments and the use of hypnotics with their episodes. These findings emphasise the critical role of stress in sleepwalking and suggest that stress management and sleep hygiene should be incorporated into treatment strategies. Furthermore, our study underscores the growing recognition of dream-like mentation as an integral component of sleepwalking in adults. Future research should focus on refining our understanding of the role of psychosocial stressors and their possible neurobiological mechanisms underlying adult somnambulism.

1 | Introduction

Sleepwalking (SW), also known as somnambulism, is a NREM sleep parasomnia that affects about 2%–4% of the general adult population and carries considerable harm potential (Hublin et al. 1997; Ingravallo et al. 2014; Mainieri et al. 2023; Ohayon et al. 1999). Although the aetiology of SW remains unclear, including the roles played by genetic, neurobiological and clinical

variables (e.g., Irfan et al. 2021; Zadra et al. 2013), several factors have been reported to increase the likelihood and intensity of somnambulist episodes in predisposed individuals.

Factors described as facilitating the occurrence of SW episodes in individuals with a predisposing genetic background have been generally grouped into priming factors (e.g., conditions that deepen or fragment sleep) and precipitating factors

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(e.g., internal and external stimuli that can trigger episodes) (Pressman 2007b, 2013). Among factors that deepen sleep, and in particular increase slow-wave sleep, sleep deprivation is the best known and most studied in sleepwalkers (Cataldi et al. 2024; Joncas et al. 2002; Pilon et al. 2008; Zadra et al. 2008), although other variables such as fever have also been investigated (Kales et al. 1979). Factors that fragment sleep and which may contribute to episode occurrence in predisposed individuals include environmental stimuli such as sudden noises (Cataldi et al. 2024; Pilon et al. 2008), the presence of stressors (Arnulf 2018; Buskova et al. 2015; Ramm et al. 2020), and comorbid sleep disorders such as obstructive sleep apnea and periodic leg movements (Espa et al. 2002; Guilleminault et al. 2003; Idir et al. 2022; Owens et al. 1997). In addition, the occurrence of somnambulistic episodes has been documented following the intake of various classes of psychotropic medications (e.g., hypnotics, antidepressants, neuroleptics) (Ben-Hamou et al. 2011; Lam et al. 2009; Pressman 2007b). These medications have been proposed to facilitate dissociation and induce SW episodes by modulating sleep and wakefulness states (Chopra et al. 2020; Dumont et al. 2024; Stallman et al. 2018).

Finally, while clinical and empirical data indicating that sleep mentation or dream-like activity can accompany somnambulistic episodes and even modulate observed behaviours during the events are not new (Oudiette et al. 2009; Zadra et al. 2013), their association with NREM parasomnias and their contributing role in how episodes unfold are being increasingly well documented (Castelnovo et al. 2024; Cataldi et al. 2024; Siclari 2024). Hence, contrary to older views in the field, dream-like experiences are now recognised as sometimes playing an integral role in how somnambulistic events are experienced by many adult sleepwalkers.

In sum, while a range of variables has been identified as potentially impacting episode occurrence in adults presenting with a history of somnambulism, their relative importance and frequency of endorsement have never been systematically investigated from sleepwalkers' perspective. Such data would not only be helpful in refining work into the disorder's aetiology, but also in the clinical assessment and understanding of potential factors that patients themselves perceive as contributing to their episodes. We thus addressed this shortcoming by investigating a broad range of self-reported priming and precipitating factors for SW in a large cohort of adults with a formal diagnosis of somnambulism.

2 | Material and Methods

2.1 | Procedure

Participants were selected among adults referred to l'Hôpital du Sacré-Cœur de Montréal's sleep disorders clinic for suspected somnambulism between 2003 and 2018. Included participants all underwent a thorough clinical assessment, which included at least one night of video-PSG, and received a final diagnosis of primary SW according to the 2nd edition of the *International Classification of Sleep Disorders* (ICSD; American Academy of Sleep Medicine 2005). As part of their assessment, participants also completed an extensive questionnaire on their history of

parasomnias, overall sleep quality, and dream recall. This questionnaire, along with the clinical assessment and video-PSG recording, was part of a larger study on the assessment and pathophysiology of adult somnambulism.

Only patients with a primary diagnosis of somnambulism were considered for the present study. The clinical assessment and overnight video-PSG were used to exclude participants with any of the following conditions: (1) presence of a neurological disorder; (2) presence of a comorbid sleep disorder, except for sleep terrors; (3) apnea index of ≥ 5 per hour of sleep or an index > 15 for respiratory events (apneas and hypopneas) or an index ≥ 10 for periodic leg movements associated with microarousals; (4) use of medications that could alter sleep EEG or architecture, motor activity during sleep, or daytime vigilance; (5) transmeridian travel or night work in the 3 months prior to the first PSG; and 6) the presence of any major psychiatric disorder.

The study was approved by the research ethics committee of the *Centre intégré universitaire de santé et de services sociaux du Nord-de-l'Île-de-Montréal—Hôpital du Sacré-Cœur de Montréal* (CIUSSS-NIM). All participants provided signed informed consent for the study.

2.2 | Self-Reported Data

During their initial clinical assessment or prior to their first overnight polysomnography, participants were asked to complete the Questionnaire on Somnambulism, Sleep and Dreams, which has been utilised in previous studies by our group (Blanchette-Carriere et al. 2024; Kalantari et al. 2022; Labelle et al. 2013). The questionnaire was developed by our research group over a 2-year period based on studies and case reports detailed in the literature in combination with material collected from sleepwalkers assessed in our sleep disorders clinic. The questionnaire was used to assess various aspects of participants' parasomnia, including the frequency of somnambulism, age of onset and family history for the disorder, frequency of episode-related sleep mentation, and potentially injurious behaviours. In addition, a wide range of potential priming and precipitating factors for SW episodes were assessed by having participants rate 24 items, including factors related to stress, sleep, dreams, environmental stimuli, and medication, on a scale from 1 (never) to 5 (always) or N/A (not applicable) for the degree to which it contributed to their experiencing episodes. The instructions for this section of the questionnaire were: 'Among the following factors, please indicate those that may trigger or increase the frequency of your SW episodes. Use the provided 1–5 scale to answer. Select 'Not Applicable' if a factor does not apply to you or if you are unsure'.

2.3 | Statistical Analyses

Frequency analyses were first carried out to assess the proportion of sleepwalkers who rated a given questionnaire item with a score of (1) never, (2) rarely, (3) sometimes, (4) often, or (5) always. The proportion of factors that participants reported as at least sometimes priming or precipitating their SW episodes was then compared to the proportion reported as never or rarely priming or precipitating episodes as a function of participant

characteristics (i.e., sex, age of onset, family history) with chi-square tests. Missing data (between 3% and 7% of participants per item) were excluded from the analysis, and a Yates' correction was not used as cell values were > 5. All analyses were performed using IBM SPSS Statistics (Version 29.0.2.0).

3 | Results

3.1 | Demographic and Clinical Profiles

Our final sample of sleepwalkers was comprised of 188 adults (114 women, 74 men) aged 18–68 years ($M=32.6$, $SD=10.1$; women: $M=32.9$, $SD=10.1$; men: $M=32.1$, $SD=10.1$). One hundred and thirteen participants (60%) reported somnambulism with a childhood onset (i.e., before the age of 12 years) and another 16% reported an onset before the age of 18 with the remaining 24% reporting an adult onset. Less than half of the participants (47%) reported a family history of SW (sample

$n=134$; 54 missing or conflicting data). One-hundred forty sleepwalkers (74%) also reported a history of sleep terrors.

3.2 | Precipitating and Priming Factors for SW

The proportion of participants who rated each of the factors as never, rarely, sometimes, often or always precipitating or priming their SW episodes is presented in Table 1. As shown in the table, psychological stress was the most frequently endorsed factor, with over a third of sleepwalkers reporting stress as ‘always’ precipitating or priming their episodes. The next most common factors reported as ‘always’ precipitating or priming episodes were ‘bad dreams’ and nightmares (15%), sleep deprivation (12%), watching an intense movie (8%), dreams (7%) and sudden noises (7%). Also of interest, only 1% of sleepwalkers reported that stress ‘never’ precipitated or primed their episodes while approximately 10% reported that dreams, including nightmares, never played a role in their episodes.

TABLE 1 | Proportion of participants who rated each of the factors as never, rarely, sometimes, often, or always precipitating or priming their SW episodes.^a

Category	Factors	Never	Rarely	Sometimes	Often	Always
Stress-related	Psychosocial stress	1%	4%	17%	41%	36%
	Watching an intense movie	21%	18%	29%	24%	8%
	Sleeping in a new environment	30%	15%	23%	26%	6%
Dream-related	Bad dreams and nightmares	9%	12%	23%	40%	15%
	Dreams	10%	19%	31%	32%	7%
Increased sleep pressure	Sleep deprivation	16%	24%	21%	27%	12%
	Irregular sleep schedules	24%	18%	25%	25%	8%
	Jetlag	46%	20%	18%	12%	4%
	Intense physical activity	49%	23%	19%	7%	2%
	Daytime sleep	59%	24%	14%	3%	0%
Endogenous and external stimuli	Need to go to the bathroom	50%	15%	18%	14%	2%
	Noisy environment	50%	18%	18%	11%	4%
	Sudden noise	35%	18%	25%	16%	7%
	Bright light	70%	15%	11%	2%	2%
	Bed partner moving abruptly	52%	12%	15%	18%	4%
	Being touched	51%	19%	17%	10%	3%
	Being too hot or cold	51%	18%	22%	6%	2%
	Fever	66%	14%	17%	1%	2%
	Pain	66%	16%	11%	5%	2%
	Migraine	66%	17%	10%	7%	0%
Substance-related	Alcohol	50%	23%	13%	12%	2%
	Caffeine	52%	21%	15%	7%	5%
	Sleeping pills	80%	7%	6%	5%	1%
	Stimulant	71%	11%	8%	9%	1%

^aThe ‘not applicable’ option was excluded from the analyses.

Using a broader range of frequency criteria, Figure 1 presents the proportion of participants who rated each of the factors as sometimes, often, or always priming or precipitating their SW episodes. Among the most frequently endorsed items were psychological stress (95%), bad dreams and nightmares (78%), dreams (70%) and sleep deprivation (60%). Factors such as watching an intense movie, irregular sleep schedules, and sleeping in a new environment were endorsed by 50% to 70% of sleepwalkers while other factors such as being touched (29%), fever (20%), pain (18%), and sleeping pills (12%) were among the least frequently endorsed.

3.3 | Precipitating and Priming Factors as a Function of Biological Sex, Age of Onset, and Family History for the Disorder

Factors reported as sometimes, often or always precipitating or priming episodes was first examined as a function of sleepwalkers' biological sex. No significant sex differences were found with the exception of 'noisy environment' and 'sleeping pills', both reported as playing a significantly greater role by a greater proportion of women than men (41% women; 21% men; $\chi^2_{(1)} = 6.133$, $p = 0.013$; 18% women; 3% men; $\chi^2_{(1)} = 4.332$, $p = 0.037$, respectively).

A similar comparison of precipitating factors between participants reporting a childhood versus adult onset for the disorder revealed that those with a childhood onset were more likely to highlight the role of bad dreams/nightmares (83% childhood onset; 62% adulthood onset; $\chi^2_{(1)} = 7.589$, $p = 0.006$) as well as dreams in general (75% childhood onset; 56% adulthood onset;

$\chi^2_{(1)} = 4.645$, $p = 0.031$) as important precipitating or priming factors. The groups did not differ on any other factors.

Finally, a comparison between sleepwalkers with versus without a family history for the disorder revealed no significant difference on any of the listed factors with the exception of 'abrupt movements of the bed partner', endorsed by a greater proportion of those with a family history of SW (44% positive history; 23% no history; $\chi^2_{(1)} = 4.874$, $p = 0.027$).

4 | Discussion

This study aimed to investigate a broad range of self-reported precipitating and priming factors for SW in one of the largest cohorts of adults with an ICD-10 diagnosis of somnambulism studied to date. The most frequently reported precipitating or priming factor for participants' episodes was stress, with an overwhelming majority (94%) of sleepwalkers indicating that this factor was at least 'sometimes' involved in their episodes and over a third (36%) reporting that it was 'always' associated with their episodes. While experimental findings demonstrating correlational or causal links between stress and SW episodes are lacking, our results align with a considerable body of descriptive and clinical literature linking situational stress (e.g., family conflicts, relationship issues, job-related stressors) to the occurrence of somnambulistic episodes in predisposed individuals (Buskova et al. 2015; Lecendreux et al. 2003; Lopez et al. 2013; Pressman 2007b). Moreover, our results suggest that even 'stressful' forms of entertainment, such as watching an intense movie, may increase the likelihood of a sleepwalker subsequently experiencing an episode. Various types of stressors

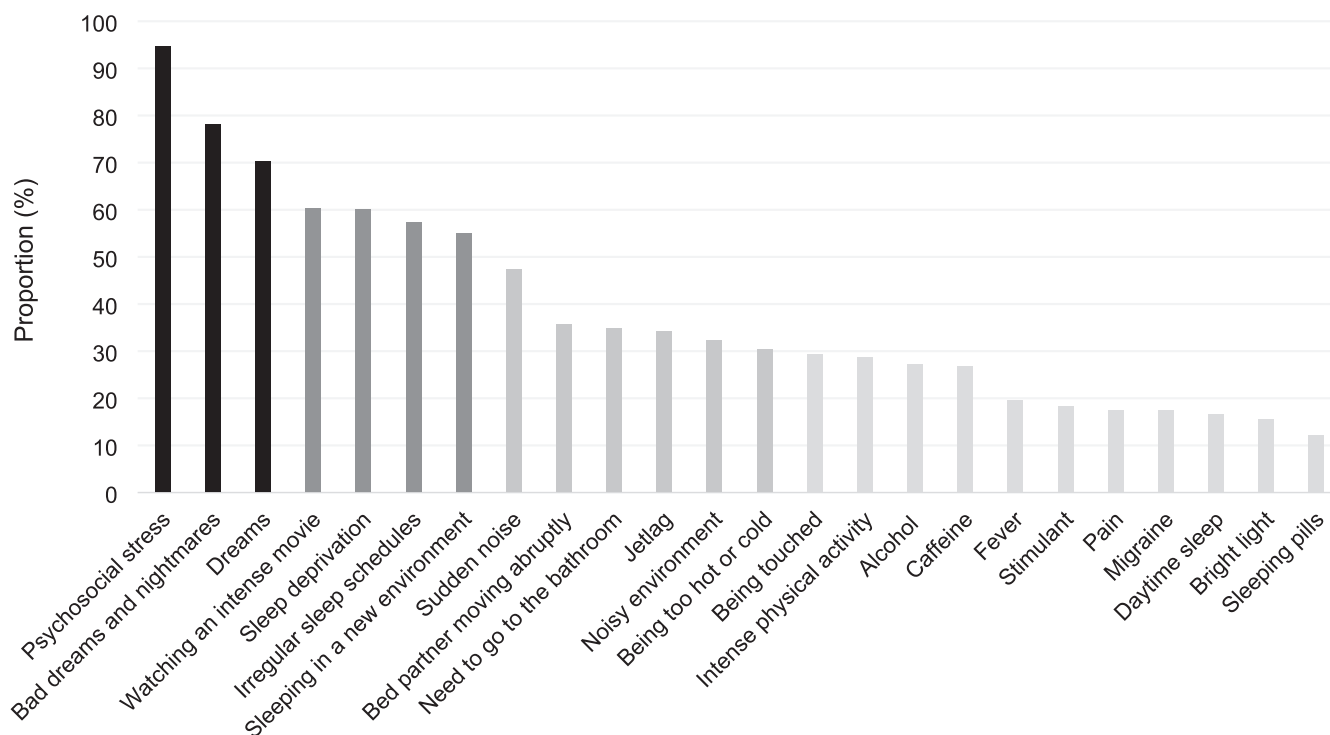


FIGURE 1 | Precipitating and priming factors for sleepwalking. Proportion of sleepwalkers who rated questionnaire items as at least sometimes precipitating or priming their episodes. Black bars indicate $\geq 70\%$ endorsement while darker grey gradients represent intermediate proportions (e.g., dark grey = 50%–<70%) and lighter tones lower proportions (e.g., light grey = 30%–<50% and lightest grey = 0%–<30%).

may prime SW episodes by increasing the fragmentation of patients' sleep, including a greater number of partial arousals from N3 sleep, which can give rise to somnambulistic behaviours (Buskova et al. 2015; Drakatos et al. 2019; Sanford et al. 2015; Smith et al. 2024). That stress is viewed by most adult sleepwalkers as playing a key role in their episodes also supports the inclusion of stress management techniques in treatment protocols for the disorder (e.g., Drakatos et al. 2019; Mundt and Baron 2021; Mundt et al. 2023).

'Bad dreams', nightmares, and dreams in general were the next most frequently reported precipitating factors for SW. While SW was long viewed as a parasomnia devoid of mental imagery or dream-like mentation, an increasing body of research indicates that adult somnambulism can, in fact, be accompanied by various forms of sleep mentation, including brief dream-like experiences and imagery that may become integrated with or superimposed onto the individual's actual physical surroundings (Castelnovo et al. 2024; Cataldi et al. 2024; Kalantari et al. 2022; Oudiette et al. 2009; Siclari 2024; Zadra et al. 2013). Given that this research highlights the often significant role of dream-like activity in shaping how somnambulistic episodes unfold and are experienced subjectively, it is unsurprising that many sleepwalkers describe their episodes as 'waking dreams' (e.g., Siclari 2024). Consequently, as shown in the present study, sleepwalkers frequently identify 'bad dreams' and nightmares as triggers for their episodes. From a clinical perspective, these findings further support a relatively new and growing body of literature that challenges the inclusion of complete amnesia for episodes and the absence of dream-like activity as defining clinical criteria for adult somnambulism (Castelnovo et al. 2024; Cordani et al. 2024; Idir et al. 2025, 2022; Lopez and Dauvilliers 2024; Zadra et al. 2013).

Among the next most frequently endorsed items were factors that either increase sleep pressure (e.g., sleep deprivation, irregular sleep schedules) or disrupt sleep continuity (e.g., sudden noises, bed partner movements). Sleep deprivation is one of the most well studied priming factors for SW, both experimentally (Blanchette-Carriere et al. 2024; Cataldi et al. 2024; Joncas et al. 2002; Zadra et al. 2008) and clinically (Buskova et al. 2015; Lopez et al. 2013; Mundt et al. 2023; Pressman 2007b). Similarly, both laboratory-based and descriptive studies have documented the role of various sleep-fragmenting or arousal-inducing factors in triggering somnambulistic episodes in sleepwalkers (Cataldi et al. 2024; Espa et al. 2002; Pilon et al. 2008; Pressman 2007a). Our findings align with this body of research, underscoring the importance of educating sleepwalkers on minimising the impact of such triggers and highlighting the need for treatment approaches to incorporate educational components that promote good sleep hygiene practices (Drakatos et al. 2019; Mundt et al. 2023).

Over 70% of sleepwalkers reported that the consumption of substances such as alcohol, caffeine, sleeping pills or stimulants was never or rarely associated with their episodes. While factors such as caffeine consumption have been anecdotally described as having potentially priming effects on episode occurrence (Cartwright and Guilleminault 2013; Moldofsky et al. 1995), our results support prior observations (e.g., Labelle et al. 2013) suggesting that these substances play a minor role in the vast

majority of episodes experienced by chronic sleepwalkers. Given longstanding debates over the lack of scientific evidence supporting an alcohol-based defence for SW-related crimes (e.g., Pressman 2019; Pressman et al. 2013; Rumbold et al. 2014), it is important to highlight that fewer than 15% of our sleepwalkers viewed alcohol as an important precipitating or priming factor for their episodes.

Turning to physical ailments such as fever, pain, or migraines, fewer than 20% of our sleepwalkers indicated that such conditions at least occasionally played a role in facilitating their episodes. The importance of these factors depends on the extent to which such conditions are generally experienced by sleepwalkers, and their relative impact may also vary across different subgroups of sleepwalkers. For example, one study found that sleepwalkers with higher levels of depression and anxiety were more likely to report pain or migraines as precipitating or priming factors for their somnambulistic episodes (Labelle et al. 2013) while another found that chronic pain was more common in older sleepwalkers who reported higher levels of insomnia, sleepiness, and depressive symptoms (Lopez et al. 2015).

Our analyses further showed that the roles attributed to the majority of the precipitating and priming factors investigated in this study did not vary significantly as a function of biological sex, age of onset, or family history for the disorder. Notably, however, a significantly greater proportion of women than men reported a 'noisy environment' and 'sleeping pills', as important precipitating factors for their episodes. One possible explanation for the former is that female sleepwalkers may be more sensitive to environmental disturbances, such as noise, due to lower arousal thresholds from NREM sleep (Won et al. 2020), making them more likely to experience noise-induced episodes. Regarding hypnotics, evidence suggests that women are more likely than men to experience insomnia and to use hypnotics across populations (e.g., Carrasco-Garrido et al. 2022; Kolla et al. 2024; Milani et al. 2021), increasing the likelihood of associating such medications with the occurrence of their episodes. Furthermore, it is well established that certain non-benzodiazepine hypnotics (e.g., zolpidem) can induce medication-related somnambulism (Stallman et al. 2018).

Finally, a few factors such as 'Sleeping in a new environment' and 'Irregular sleep schedules' were just as likely to be reported as 'often' priming participants' episodes as 'never' doing so. This suggests, for instance, that while sleeping in a novel environment may act as a priming factor in some sleepwalkers, it may be protective in others. These observations highlight the importance of investigating individual differences and patient subgroups to better understand the differential effects of various priming factors associated with SW.

Certain limitations must be considered when interpreting the present findings. First, our results may not be generalizable to sleepwalkers in the general population as adult sleepwalkers referred to a sleep disorders clinic typically present with a more severe and/or frequent form of the disorder. Similarly, since only adult sleepwalkers were included in the study, the results do not necessarily extend to children and young adolescents in which other precipitating and priming factors may play greater or different roles. Second, some factors with low endorsement

levels are closely tied to participants' exposure to them, which was not measured in the present study. For instance, people who rarely travel across time zones are unlikely to report 'jet-lag' as a precipitating factor for their SW just like 'intense physical exercise' can only be reported by people who have engaged in it. Moreover, given that sleepwalkers may experience variable levels of amnesia for their episodes, some episodes as well as the factors that facilitated their occurrence may have gone unnoticed.

5 | Conclusion

This study provides one of the most comprehensive examinations to date of self-reported precipitating factors for SW in adults with a formal diagnosis of somnambulism. Stress emerged as the most frequently reported trigger, reinforcing the need to integrate stress management strategies into treatment protocols. Further research is needed to better characterise the nature of psychosocial stressors most likely to prime or precipitate SW in predisposed individuals and to elucidate the psychological and neurobiological mechanisms through which these effects may operate. Additionally, the findings highlight the role of dream-related mentation in SW episodes, lending further support to recent research challenging traditional clinical criteria for the disorder. Factors that increase sleep pressure or disrupt sleep continuity were also commonly endorsed, emphasising the importance of promoting good sleep hygiene among sleepwalkers. In contrast, substance use and physical ailments were generally perceived as minor contributors, although individual variability may exist based on comorbid conditions or subgroup differences. Finally, while most reported precipitating and priming factors were consistent across demographic variables, the observed sex differences in sensitivity to environmental noise and hypnotic use suggest potential avenues for future investigation. Collectively, these findings enhance our understanding of SW triggers and provide valuable insights for improving both clinical interventions and patient education.

Author Contributions

Cloé Blanchette-Carrière: data curation, formal analysis, visualization, project administration, writing – original draft, writing – review and editing. **Mathieu Pilon:** conceptualization, methodology, writing – original draft, writing – review and editing. **Simon Trudeau:** data curation, visualization, project administration, writing – review and editing. **Lydia Hébert-Tremblay:** investigation, data curation, writing – original draft. **Alex Desautels:** conceptualization, investigation, supervision, funding acquisition, writing – review and editing. **Jacques Montplaisir:** conceptualization, investigation, supervision, funding acquisition, writing – review and editing. **Antonio Zadra:** conceptualization, methodology, data curation, funding acquisition, supervision, writing – original draft, writing – review and editing.

Conflicts of Interest

A.D. received operating grants from CHIR, AASM, CHSF and research grants from Eisai and Takeda; honoraria from serving on the scientific advisory board of Eisai, Paladin Labs, as well as honoraria from speaking engagements from Eisai, Jazz Pharma, Paladin Labs and AstraZeneca. None of the financial disclosures are relevant to the submitted work. Other authors report no conflicts of interest related to this article.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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